



STEP-BY-STEP MIGRATION GUIDE

VMware to *Apache* *CloudStack*



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About this Guide

This guide is designed for VMware users who want to migrate their existing vSphere environment to a robust IaaS cloud environment based on Apache CloudStack and the KVM Hypervisor, ensuring a smooth, low-friction migration journey.

Upon completion of this guide, users will emerge with a comprehensive understanding and practical know-how, ready to manage a versatile cloud infrastructure with CloudStack and able to seamlessly handle an environment fully managed by CloudStack's unified UI or API.

Why Migrate from VMware to Apache CloudStack

Introduction to Apache CloudStack

Apache CloudStack stands out for its comprehensive features that meet the needs of modern cloud computing. Below are some outstanding features:

- **Multi-hypervisor support:** CloudStack is compatible with various hypervisors, allowing for a flexible infrastructure.
- **Multi-tenant architecture:** It ensures that multiple users or organisations can securely share the same physical infrastructure while maintaining privacy and security.
- **High availability:** CloudStack is designed to minimise downtime, ensuring that services are always accessible.
- **Simplified orchestration:** It streamlines the management of compute, network, and storage components, making it easier for organizations to manage their cloud resources.
- **Fully open-source solution:** Apache CloudStack is an open-source solution, powered by an active community. Typically, there are 2 major releases per year and the project roadmap follows the needs and requirements of its users.
- **Free and ensuring your business profitability:** CloudStack is a free to use solution, and it ensures your profitability in the long-term.

Benefits of Migrating to Apache CloudStack

The migration from VMware vSphere to Apache CloudStack is considered as a strategic decision for several reasons:

- **Open-source platform:** CloudStack is an open-source cloud orchestration platform, known for its adaptability and efficiency in managing large-scale cloud environments.
- **Versatility:** It supports any underlying hardware compatible with the hypervisor, offering organisations the flexibility to optimize their cloud management strategies.
- **Proven reliability:** CloudStack is trusted and utilized by the world's largest cloud environments, demonstrating its capability to handle extensive cloud operations. It is proven at large-scale production deployments, scales easily and ensures multi-tenancy.

The Impact of VMware Acquisition by Broadcom

The acquisition of VMware by Broadcom has led to significant changes, prompting organizations to seek alternatives:

- **End of Availability (EOA) for products and partner programs:** This has created a need for a stable and reliable alternative solutions in the long-term.
- **Changes in licensing models and VMware pricing:** Organisations are looking for cost-effective and flexible cloud solutions.
- **Seeking robust alternatives:** With the shift in VMware's strategy, Cloud Service Providers (CSPs), Managed Service Providers (MSPs), and enterprises are exploring robust alternatives like Apache CloudStack.

[Broadcom's acquisition of VMware has led to major strategic changes in VMware's operating structure. Read more here.](#)

Apache CloudStack and VMware Side-by-Side Comparison

Apache CloudStack offers several advantages and way better flexibility over VMware, making it an attractive option for organisations:

- **Open-source and hypervisor-agnostic:** Unlike VMware, CloudStack is open-source and not tied to a specific hypervisor, offering greater flexibility.
- **Feature-rich comparable to VMware:** CloudStack provides a range of features that are on par with VMware's offerings but with the added benefits of open-source flexibility.

For a detailed comparison between VMware and Apache CloudStack, including features and capabilities, visit the following link: [VMware Alternatives Comparison Matrix](#). This comparison provides insights into how CloudStack stands as an alternative to VMware, especially in the wake of changes brought about by Broadcom's acquisition.

VMware & CloudStack Side-by-Side Comparison - Feature Matrix

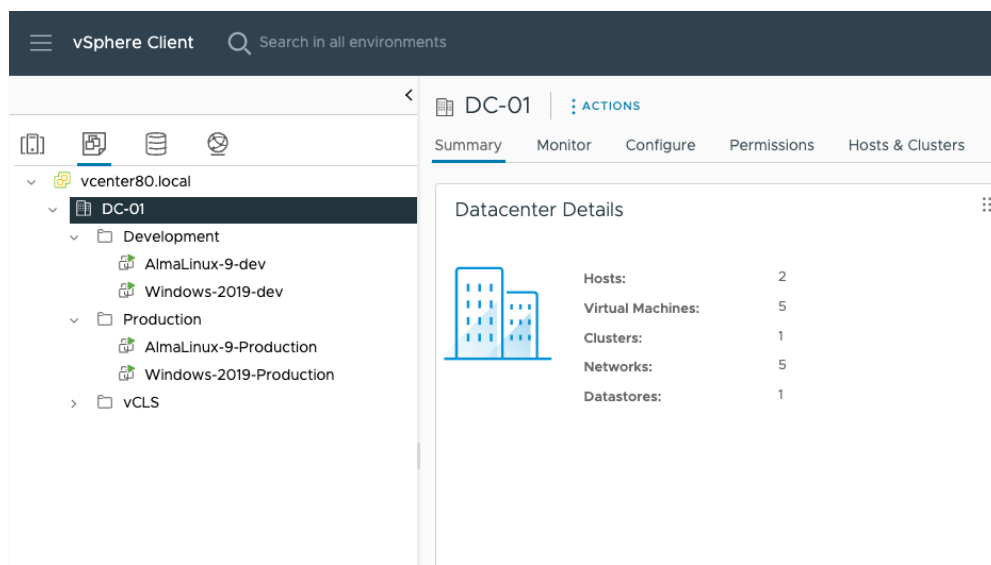
FEATURE	vSphere Essential Plus*	vSphere Standard*	vSphere Foundation*	VMware Cloud Foundation	CloudStack + KVM
Multi-hypervisor Support	No	No	No	No	Yes
Multi-zone Management	No	No	No	Yes	Yes
DRS	No	No	Yes	Yes	Yes
Distributed Switch	No	No	Yes	Yes	Partial
Cross Cluster live-migration	No	No	Yes	Yes	Yes
Cross DC live-migration	No	No	Yes	Yes	Partial
Storage live-migration	No	No	Yes	Yes	Yes
High Availability	Yes	Yes	Yes	Yes	Yes
Kubernetes Support	No	No	Yes	Yes	Yes
Self-Service	No	No	No	Yes	Yes
Disaster Recovery	No	No	Add-on	Add-on	Partial
SDN Integration	No	No	No	Yes	Yes
SDS Integration	No	No	Yes	Yes	Yes
Multi-Cloud	No	No	No	Yes	No
Hybrid Cloud	No	No	Yes	Yes	Yes
Capacity Analytics	No	No	Yes	Yes	Yes
Life-cycle Management	No	No	No	Yes	Yes
Firewall	No	No	Add-on	Add-on	Yes
Load Balancer	No	No	Add-on	Add-on	Yes
Usage Metering	No	No	No	Yes	Yes
Object Storage	No	No	Yes	Yes	Yes
Edge Zone Capabilities	No	No	No	Yes	Yes
VNF Orchestration	No	No	No	No	Yes

* The combination of CloudStack and vSphere is also possible. In this case, the missed features will be included if the use of vSphere is considered.

Example of a Migration Scenario

In this Guide, we will delve into a practical migration scenario to illustrate the transition process from VMware to Apache CloudStack. Our example will focus on a vSphere cluster configuration that includes 2 ESXi hosts. Within this cluster, we manage a total of 4 virtual machines (VMs), which consist of 2 Windows and 2 Linux Instances divided into 2 environments, development, and production.

This scenario is obviously much smaller than most production environments, but it allows the demonstration of the key migration concepts in this guide. Those concepts can easily be extrapolated to run a much larger migration.



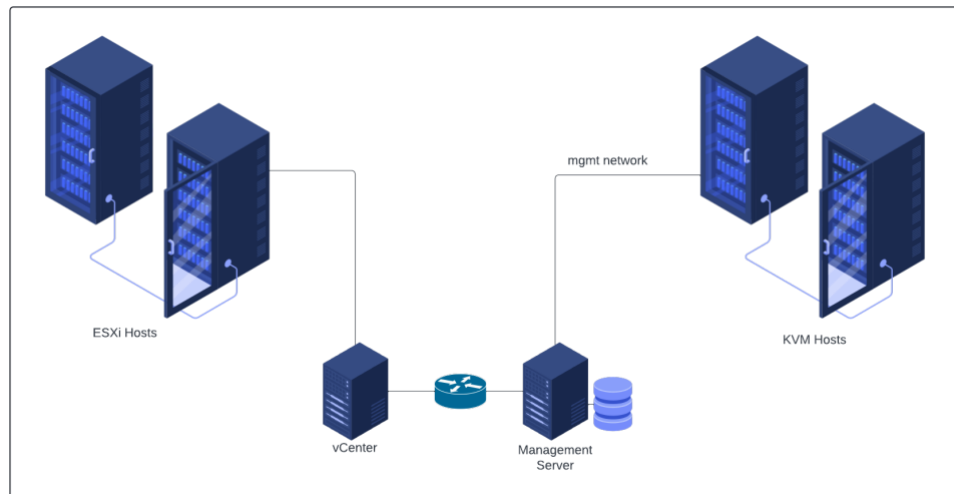
These are the VM details and their respective environments and networks:

Virtual Machine	Environment	vCPUs	Memory	Storage	Network
AlmaLinux-9-dev	Development	2	2GB	20GB	Development-Network
Windows-2019-dev	Development	4	4GB	100GB	Development-Network
AlmaLinux-9-production	Production	2	2GB	20GB	Production-Network
Windows-2019-production	Production	4	4GB	100GB	Production-Network

Migration Design

When planning a migration from a VMware-based infrastructure to Apache CloudStack and KVM hypervisor, the first step is to outline a design architecture. This design will illustrate the placement and interaction of key components in the network, ensuring that everything happens as expected during migration. The main components to consider in this architecture are vCenter, CloudStack Management Server, and KVM hosts.

Components Placement and Network Design



vCenter

During the initial migration phase, the CloudStack Management Server will establish a connection to vCenter to retrieve the list of Instances hosted on ESXi hosts. For this process to proceed smoothly and quickly, the vCenter must be accessible to the CloudStack Management Server within a Layer 2 (L2) or Layer 3 (L3) network segment in the same data centre infrastructure, to ensure greater copy speed of the Instance disks between the vCenter and the CloudStack Storages.

Management Server

The CloudStack Management Server is the heart of the Apache CloudStack environment, responsible for orchestrating cloud resources, managing the cloud infrastructure, and interfacing with the user. In the network architecture, the Management Server should be positioned for optimal connectivity to both the KVM hosts and the vCenter. This ensures that the Management Server can effectively manage the migration process and communicate with all parts of the infrastructure.

KVM Hosts

KVM hosts are the foundation of the new cloud infrastructure, responsible for running Instances and delivering compute resources. In terms of network architecture, KVM hosts must be deployed to improve performance and resilience. The Proof of Concept (PoC) Guide details the configuration of the network to ensure that these objectives are covered.

Migration Plan

Planning your migration from VMware to Apache CloudStack is a process that involves several steps to ensure a smooth transition:

Planning Your Migration Journey

The suggested approach to migration assumes the gradual transition from VMware vSphere to the KVM hypervisor. This strategy is minimising risk, leveraging available resources and meeting the overall business objectives of the migration project.

The migration journey will follow these steps:

- Installing and configuring Apache CloudStack
- Configuring KVM Hosts
- Preparing the CloudStack Environment
 - Domain and Accounts
 - Service Offerings
 - Networks
- Migrating Instances

Profile the VMware Instances to Migrate

Performing an analysis of your VMware infrastructure is essential before migrating to Apache CloudStack. Evaluate key aspects such as the number of virtual CPUs, memory allocation, disk usage, configuration complexity, and dependencies of other services. This comprehensive assessment helps you understand the technical landscape, identify potential challenges, and ensure a well-established migration plan that meets specific needs and minimises disruption to operations.

Migration Timeline

Developing a detailed migration schedule is helping you to have a smooth and easy migration. Include milestones and deadlines, allocating time for testing, troubleshooting, and transitioning workloads.

Migration Automation

This guide focusses on explaining the key concepts and using tooling that is part of the Apache CloudStack GUI. When undertaking a migration at scale, most organisations will choose to automate the process via the CloudStack API or CLI.

Installing and Configuring Apache CloudStack

Deploying Apache CloudStack

If you are unfamiliar with CloudStack, we recommend to download and explore the CloudStack [Proof-of-Concept Guide](#). It will help you to get a comprehensive understanding of installation and management, facilitating a seamless transition.

Configuring KVM Hosts

To start migrating from vSphere to KVM, at least one CloudStack-managed KVM cluster shall be available, with a minimum of two Hosts to accommodate the initial workload. These hosts will be used for the conversion process and to accommodate the migrated Instances. As the migration progresses and VMware Instances are migrated to the KVM hypervisor, thus freeing an ESXi host, you can remove that host from the vSphere cluster. Then proceed with the installation of KVM and integrate it into the existing cluster through the CloudStack UI, ensuring that there are at most 16 Hosts per cluster.

Supported Linux Distributions for KVM Hosts

Below is a table of the supported Linux distributions for KVM hosts:

Linux Distribution	Supported Versions
Alma Linux	8, 9
Red Hat Enterprise Linux	8, 9
Rocky Linux	8, 9
Ubuntu	22.04 LTS

Migration Tools Components

Virt-v2v Tool

The cloudstack-agent package, installed into KVM Hosts, does not automatically include the installation of the virt-v2v package. Virt-v2v is essential for converting VMware Virtual Machines into the KVM format. Therefore, it is necessary to manually install the virt-v2v package on all KVM Hosts to enable this functionality. Due to EL9 hosts (e.g. Alma Linux 9) having newer version of virt-v2v tool, we suggest doing the conversion on EL9 hosts, for a greater success rate.

To install virt-v2v on Red Hat Enterprise Linux, Rocky Linux, or Alma Linux, open the terminal in each KVM Host and execute the following command:

```
dnf -y install virt-v2v
```

For Ubuntu Linux distribution, execute the following command to install virt-v2v, additionally installing nbdkit package:

```
apt -y install virt-v2v nbdkit
```

Virtio Drivers for Windows Guest Instances

To convert Windows Guest Instances, install the virtio-win package to provide the Windows virtio drivers on KVM hosts, as it is required for virt-v2v.

For Red Hat Enterprise Linux, Rocky Linux, or Alma Linux distributions, execute the following command:

```
dnf -y install https://fedorapeople.org/groups/virt/virtio-win/direct-downloads/stable-virtio/virtio-win.noarch.rpm
```

For Ubuntu Linux Distribution, execute the following command:

```
apt -y install virtio-win
```

Preparing the CloudStack Environment

This section of the guide focuses on preparing CloudStack for the migration of your VMware Instances to CloudStack. CloudStack preparation involves setting up a structured environment, including Domain, Accounts, Service Offerings and Networks.

Domain and Accounts

Before starting the migration, plan and configure a multi-tenancy framework that mirrors or enhances the existing VMware configuration. CloudStack's multi-tenancy capabilities are managed through Domains and Accounts, providing a tiered approach to resource management and access control.

Domains

Think of Domain as organisational units within CloudStack that enable hierarchical resource management. Domains can represent different departments, projects, or even separate customers in a service provider model. Each Domain can contain multiple Accounts and provides scope for policies, resource limits, and permissions.

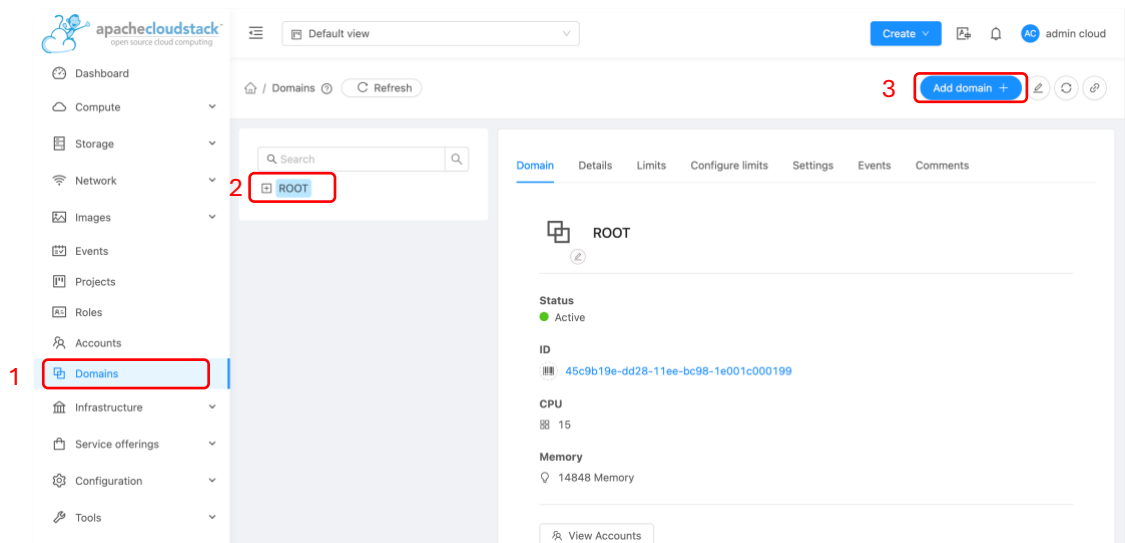
Accounts

Within Domains, Accounts serve as primary entities to manage resources and deploy Instances. Each Account can have a set of virtual resources, including Compute Instances, Networks, and Storage Volumes. When setting up Accounts, consider the resource limits and permissions needed to align with your organisation's operational and security policies.

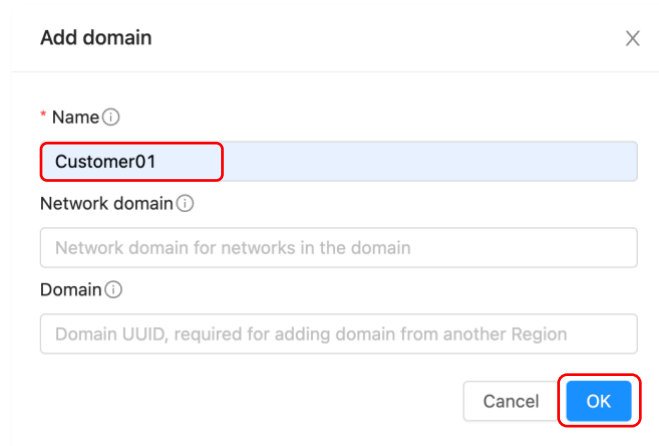
Creating the Domain and Accounts Structure

Following the suggested example, we need to create 1 Domain (Customer01) and 2 Accounts within the Domain (development and production).

1. In the CloudStack UI, log in as admin, click on “Domains” in the side menu bar, click on the ROOT Domain, and then click on the “Add domain” button:



2. In the form, enter the appropriate name for the Domain. Repeat this process as many times as necessary for each tenant you identified in your VMware infrastructure. In this case, we are creating the Domain “Customer01” following the example.



Add domain

* Name ⓘ

Customer01

Network domain ⓘ

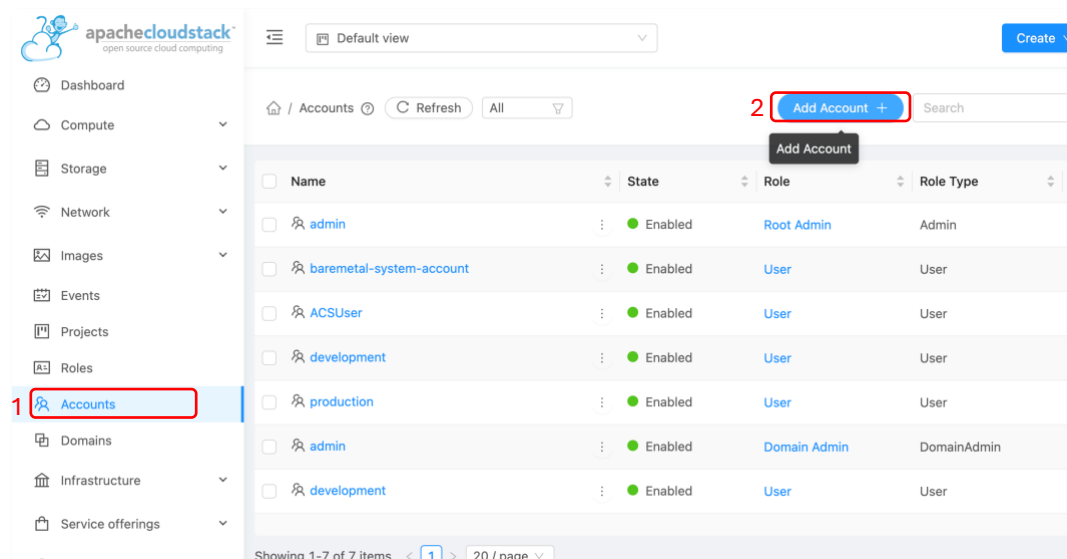
Network domain for networks in the domain

Domain ⓘ

Domain UUID, required for adding domain from another Region

Cancel OK

3. Now, navigate to “Accounts” in the sidebar menu and then click on the “Add Account” button:



apachecloudstack™ open source cloud computing

Dashboard

Compute

Storage

Network

Images

Events

Projects

Roles

1 Accounts

Domains

Infrastructure

Service offerings

Default view

Create

2 Add Account +

Add Account

Name	State	Role	Role Type
☐ admin	Enabled	Root Admin	Admin
☐ baremetal-system-account	Enabled	User	User
☐ ACSUser	Enabled	User	User
☐ development	Enabled	User	User
☐ production	Enabled	User	User
☐ admin	Enabled	Domain Admin	DomainAdmin
☐ development	Enabled	User	User

Showing 1-7 of 7 items < 1 > 20 / page

4. Fill it in with the appropriate information for the Account. In this example, we are creating both Accounts, “development” and “production”, within the Domain “Customer01” where the role is “User”. In this form, the user dev-user and prod-user will be created, as an Account need to have at least one user. You can add more users once the Account is created.

The image shows two side-by-side 'Add Account' forms. The left form is for a 'development' account and the right form is for a 'production' account. Both forms are for the 'Customer01' domain and the 'User' role. The forms include fields for Username, Password, Confirm password, Email, First name, Last name, Domain, Account, Timezone, and Network domain. Arrows point from labels to specific fields: 'First user within Account' points to the Username field in the development form, and 'Account information' points to the Role, Username, Password, Confirm password, Email, First name, Last name, and Account fields in the production form.

First user within Account

Account information

Left Form (development):

- Role: User (User)
- Username: dev-user
- Password: [masked]
- Confirm password: [masked]
- Email: dev-user@customer-1
- First name: Developer
- Last name: User
- Domain: ROOT/Customer01
- Account: development
- Timezone: CET [Central European Time]
- Network domain: Network domain for the account's networks

Right Form (production):

- Role: User (User)
- Username: production-user
- Password: [masked]
- Confirm password: [masked]
- Email: prod-user@customer-1
- First name: Production
- Last name: User
- Domain: ROOT/Customer01
- Account: production
- Timezone: CET [Central European Time]
- Network domain: Network domain for the account's networks

Service Offerings

The next step is to ensure that your CloudStack environment can provide the computing resources you need for your migrated Instances. This involves creating Service Offerings in CloudStack that match with the VMware Instances configurations.

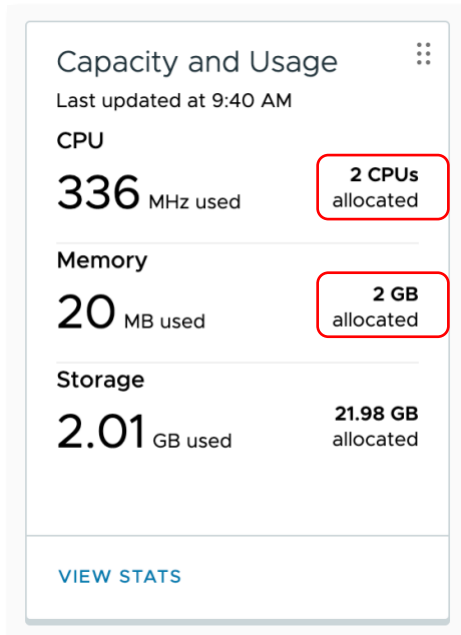
Compute Offerings

Identifying Compute Offerings in vCenter

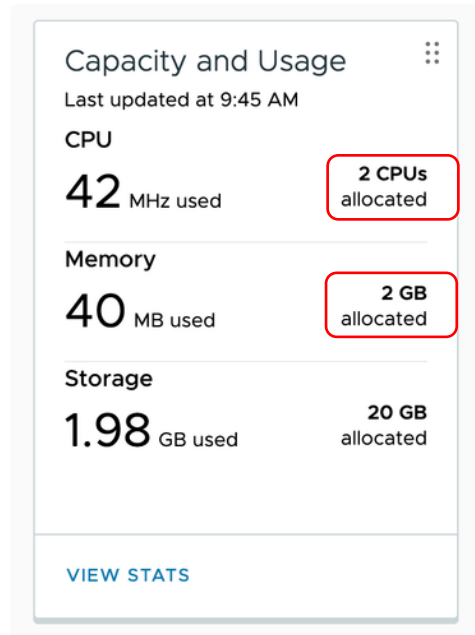
Start cataloguing resource allocations (CPU, memory, storage, networking capacity) of Instances in your VMware environment. Identify common configurations to simplify the creation of CloudStack Compute Offerings. For each group of VMware Instances with similar resource allocations, create a related Compute Offering on CloudStack. Ensure the Compute Offering is aligned with VMware Instances CPU and memory specifications to ensure performance parity.

Below we list 2 Compute Offerings according to the suggested example extracted from vCenter:

- 2 vCPUs and 2GB of Memory

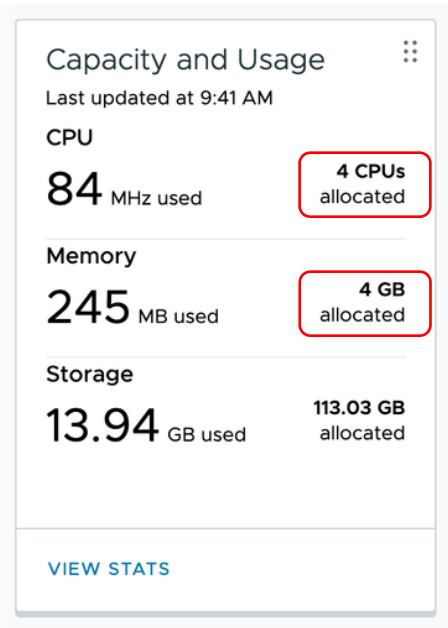


AlmaLinux-9-Dev

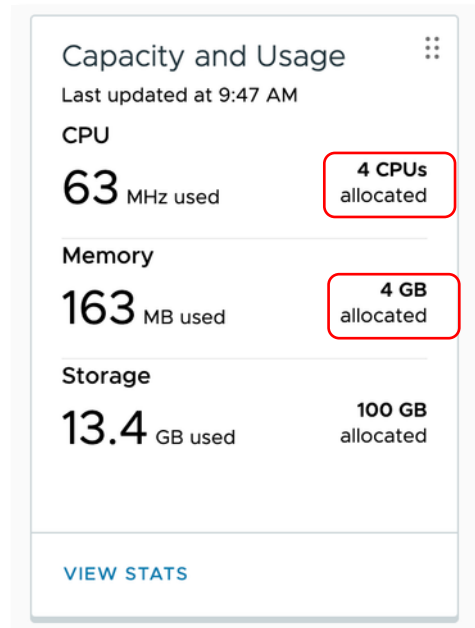


AlmaLinux-9-Production

- 4 vCPUs and 4GB of Memory



Windows-2019-Dev

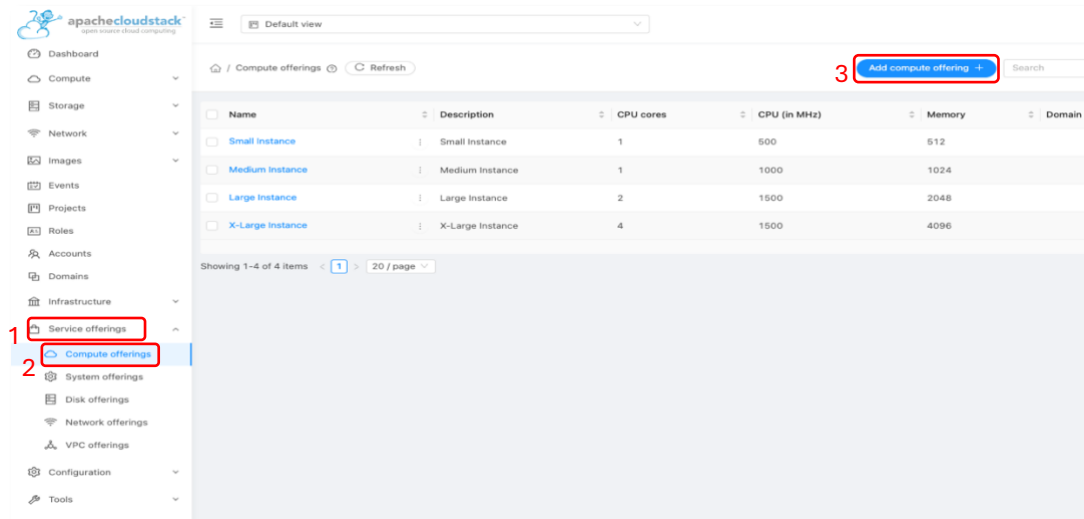


Windows-2019-Production

Creating Compute Offerings

With the Instance characteristics grouped, create the appropriate Computing Offerings for each of them.

1. In the CloudStack UI, log in as admin, click on “Service Offerings” > “Compute Offerings” in the side menu bar and then click on the “Add Compute Offering” button:



2. It will open a form as follows. Fill it in following the Instances profiles you identified in vCenter.

Add compute offering ⓘ

* Name ⓘ

Description ⓘ

Compute offering type

Fixed offering

Custom constrained

Custom unconstrained

* CPU cores ⓘ

the CPU number of the servi...

* CPU (in MHz) ⓘ

the CPU speed of the service...

* Memory (in MB) ⓘ

the total memory of the servi...

Host tags ⓘ

the host tag for this service offering.

Network rate (Mb/s) ⓘ

data transfer rate in megabits per second allow...

Offer HA ⓘ

Dynamic scaling enabled ⓘ

CPU cap ⓘ

Volatile ⓘ

Deployment planner ⓘ

GPU

None

Public

Zone ⓘ

the ID of the containing zone(s), null for public offerings

Compute only disk offering ⓘ

Storage type ⓘ

Shared

Local

Provisioning type ⓘ

Thin provisioning

Sparse provisioning

Fat provisioning

Write-cache type ⓘ

No disk cache

Write-back disk caching

Write-through

QoS type

None

Hypervisor

Storage

Root disk size (GB) ⓘ

the Root disk size in GB.

Storage tags ⓘ

the tags for this service offering.

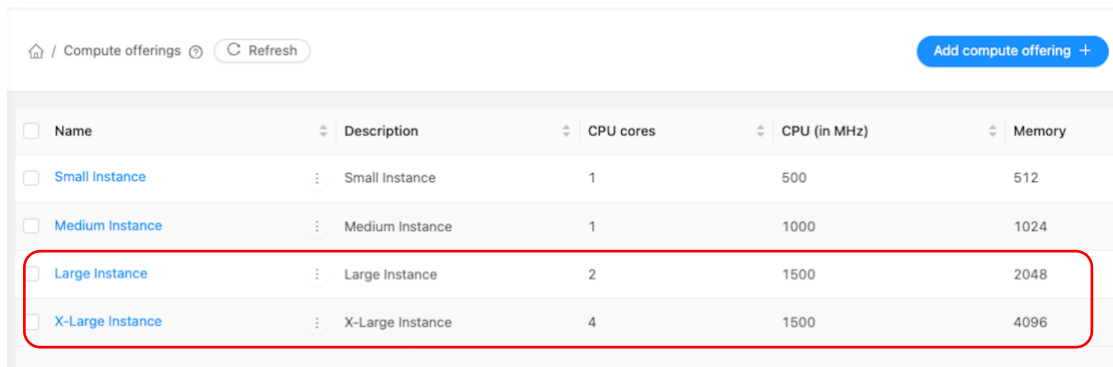
Encrypt ⓘ

Disk offering strictness ⓘ

Cancel

OK

- For this example, we have created 2 Computing Offerings, one for the Instance Profile “2 vCPUs and 2GB of Memory” and another for “4 vCPUs and 4GB of Memory”.



☐	Name	Description	CPU cores	CPU (in MHz)	Memory
☐	Small Instance	Small Instance	1	500	512
☐	Medium Instance	Medium Instance	1	1000	1024
☐	Large Instance	Large Instance	2	1500	2048
☐	X-Large Instance	X-Large Instance	4	1500	4096

Optional: Disk and Network Offerings

Disk Offering

If your VMware Instances have specific storage requirements, create specific Disk Offerings in CloudStack that meet those needs. This may include specifying storage size and performance characteristics such as throughput and IOPS.

Network Offering

While not directly related to the Instances configurations, ensure that your CloudStack Network Offerings can support the bandwidth requirements and network services (firewalls, load balancers, network throughput, etc) used by your VMware Instances.

Networks

Before you start migrating your VMware Instances to CloudStack, create Networks within CloudStack. These networks must be configured under the appropriate Accounts. During the migration process, you can associate each Instance to the correct Network. This configuration will ensure connectivity, isolation, and security for your migrated Instances.

CloudStack supports multiple network types, each serving different use cases:

- Isolated Networks**
Ideal for environments where isolation between different workloads is a must. Consider setting up Isolated Networks for applications or workloads that require strict security limits.
- VPC Networks**
Virtual private cloud (VPC) Networks offer enhanced control and isolation, similar to Isolated Networks, but with the added flexibility of defining multiple subnets and more complex routing rules. VPC Networks are suitable for configuring a more structured network architecture that mirrors corporate network configurations.

- **Shared Networks**

If the migration scenario involves workloads that can share network resources without compromising security or performance, Shared Networks will provide a simplified configuration. Shared Networks can also be beneficial in providing Internet access or common services to a group of Instances.

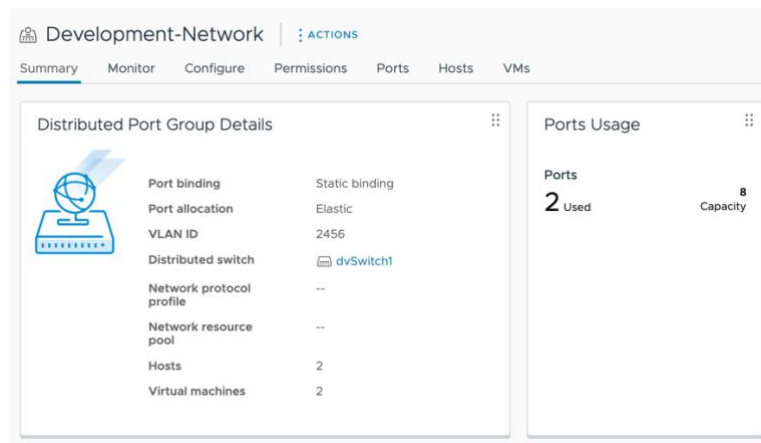
- **L2 Networks**

For workloads that require direct access to Layer 2 networks or that have specific network requirements not covered by top-level services, L2 Networks offer maximum control. This type can be used for specialised or legacy applications with specific VLAN requirements.

Identifying Networks in vCenter

In vCenter, identify the networks and which Instances they are associated with. According to the suggested example, two Networks will need to be present on CloudStack each with a related Account.

- Development-Network (within the “development” Account)

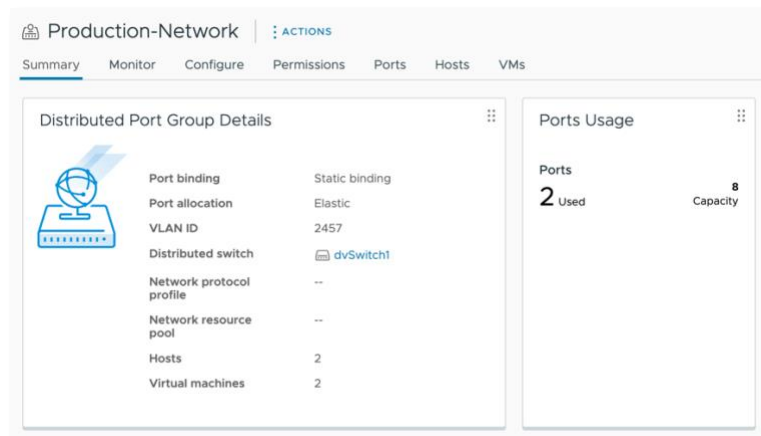


The screenshot shows the vCenter configuration page for a network named "Development-Network". The page has a top navigation bar with tabs: Summary, Monitor, Configure, Permissions, Ports, Hosts, and VMs. The "Summary" tab is selected. The main content area is divided into two panels. The left panel, titled "Distributed Port Group Details", contains a table of network settings: Port binding (Static binding), Port allocation (Elastic), VLAN ID (2456), Distributed switch (dvSwitch1), Network protocol profile (--), Network resource pool (--), Hosts (2), and Virtual machines (2). The right panel, titled "Ports Usage", shows a bar chart with "Ports" on the y-axis, "Used" at 2, and "Capacity" at 8.

Property	Value
Port binding	Static binding
Port allocation	Elastic
VLAN ID	2456
Distributed switch	dvSwitch1
Network protocol profile	--
Network resource pool	--
Hosts	2
Virtual machines	2

Ports Usage: 2 Used, 8 Capacity

- Production-Network (within the “production” Account)



The screenshot shows the vCenter configuration page for a network named "Production-Network". The page has a top navigation bar with tabs: Summary, Monitor, Configure, Permissions, Ports, Hosts, and VMs. The "Summary" tab is selected. The main content area is divided into two panels. The left panel, titled "Distributed Port Group Details", contains a table of network settings: Port binding (Static binding), Port allocation (Elastic), VLAN ID (2457), Distributed switch (dvSwitch1), Network protocol profile (--), Network resource pool (--), Hosts (2), and Virtual machines (2). The right panel, titled "Ports Usage", shows a bar chart with "Ports" on the y-axis, "Used" at 2, and "Capacity" at 8.

Property	Value
Port binding	Static binding
Port allocation	Elastic
VLAN ID	2457
Distributed switch	dvSwitch1
Network protocol profile	--
Network resource pool	--
Hosts	2
Virtual machines	2

Ports Usage: 2 Used, 8 Capacity

Creating the Networks within Accounts

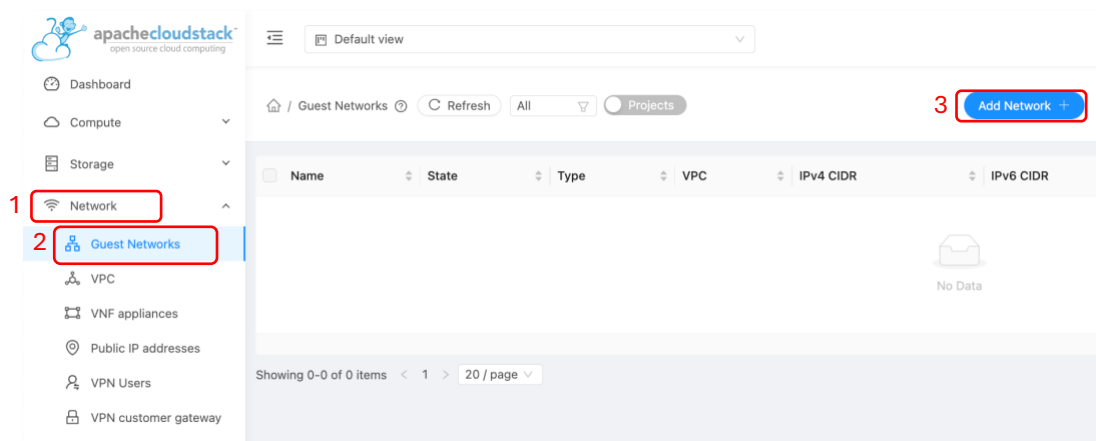
With the Networks identified in vCenter, we will now create the Networks for each Account in CloudStack.

1. In the CloudStack UI, log in to the development account using the following credential or the related Account mapped in your setup:

Username: dev-user
Domain: Customer01



2. Click on "Network" > "Guest Networks" in the side menu bar and then click on the "Add Network" button. In this case, we will create an Isolated Network as proposed by the suggested example:



- Fill it in the form with the Network Name and any other parameter in accordance with each Network's needs and then, click OK. In this case, we will create the "Development-network" following the example:

The screenshot shows the 'Add Network' dialog box with the following fields and values:

- Name:** Development-Network
- Description:** the display text of the network
- Zone:** London-Zone
- Network domain:** network domain
- Network offering:** Offering for isolated networks with Source Nat service enabled
- External Id:** ID of the network in an external system.
- Gateway:** the gateway of the network. Required for shared networks and isolated networks when L...
- Netmask:** the netmask of the network. Required for shared networks and isolated networks when L...
- DNS 1:** the first IPv4 DNS for the network
- DNS 2:** the second IPv4 DNS for the network
- IPv4 address for the VR in this Network:** IPv4 address to be assigned to the public interface of the network router. This address ...
- Network domain:** network domain

Buttons: Cancel, OK

- Log out the dev-user and log in to the production Account using the following credential or the related Account mapped in your setup:

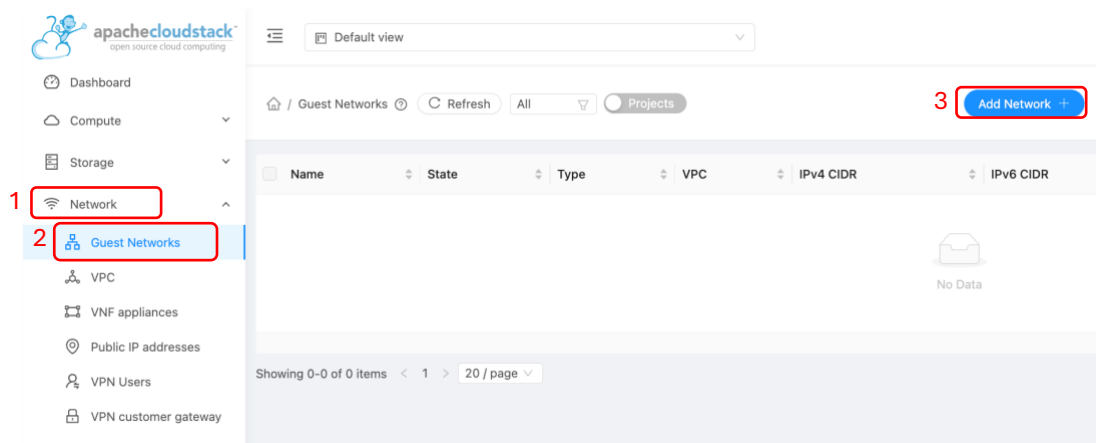
Username: prod-user
Domain: Customer01

The screenshot shows the Apache CloudStack login page with the following fields and values:

- Username:** prod-user
- Password:** masked with dots
- Domain:** Customer01

Buttons: Login

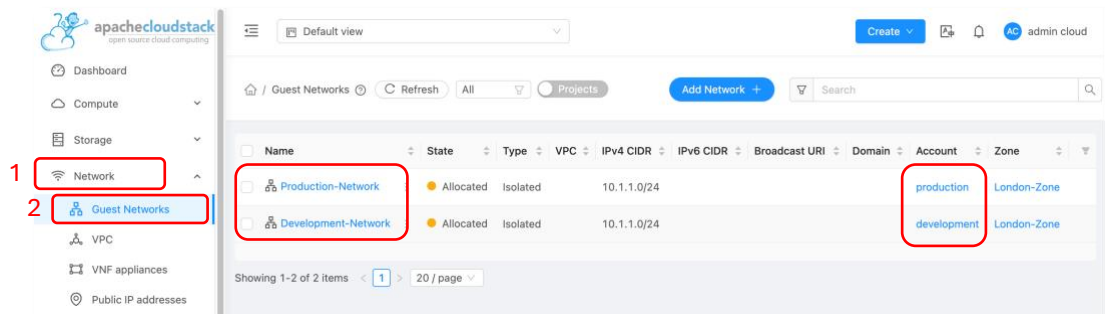
- Click on “Network” > “Guest Networks” in the side menu bar and then click on the “Add Network” button. In this case, we will create an Isolated Network as proposed by the suggested example:



- Fill it in the form with the Network Name and any other parameter in accordance with each Network’s needs and then, click OK. In this case, we will create the “Production-Network” following the example:

The screenshot shows the 'Add Network' form in the Apache CloudStack interface. The form is titled 'Add Network' and has a close button in the top right corner. It contains several fields and options for creating a new network. The 'Isolated' tab is selected. The fields are: Name (Production-Network), Description (the display text of the network), Zone (London-Zone), Network domain (network domain), Network offering (Offering for Isolated networks with Source Nat service enabled), External Id (ID of the network in an external system), Gateway (the gateway of the network. Required for shared networks and isolated networks when I...), Netmask (the netmask of the network. Required for shared networks and isolated networks when I...), DNS 1 (the first IPv4 DNS for the network), DNS 2 (the second IPv4 DNS for the network), IPv4 address for the VR in this Network (IPv4 address to be assigned to the public interface of the network router. This address ...), and Network domain (network domain). The form has 'Cancel' and 'OK' buttons at the bottom right.

7. Log out and log in as admin and then click on “Network” > “Guest Networks”. It will show the created Network for both Accounts as follows:



Migrating Instances

How CloudStack Migration Tool Works

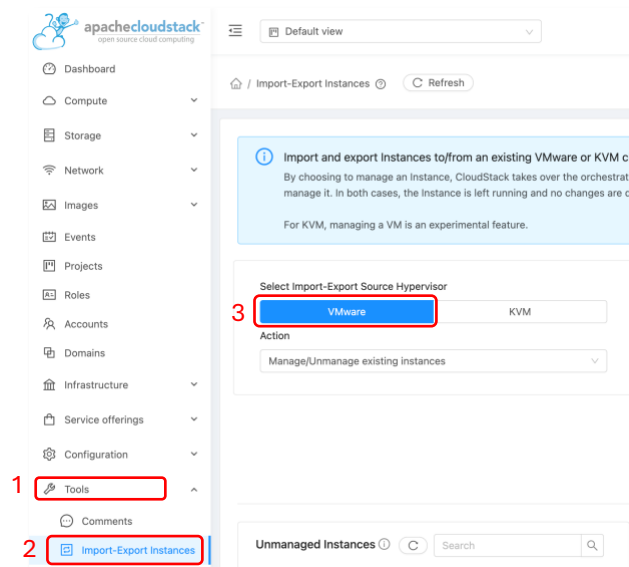
Starting with CloudStack 4.19, an automation migration tool was introduced to speed up the migration of Instances from VMware to CloudStack running KVM Hypervisor. In essence, this tool leverages the *virt-v2v* utility, which should be present on KVM hosts, to facilitate the migration process. When a user selects a VMware Instance for migration, the tool initiates a sequence of operations that begin with a connection to vCenter. It performs a disk copy operation and after transfers the copied disk to the CloudStack Secondary Storage, which is the default option, or to the Primary Storage, if the user specifies.

After copying, it converts the disk from OVA to QCOW2 format, ensuring compatibility with KVM. In addition, *virt-v2v* also automates the removal of VMware-specific drivers and the installation of KVM drivers. This is complemented by tweaks to the operating system to ensure that the migrated Instance runs smoothly in the new environment. Finally, the Instance is deployed to the KVM host and is ready for use. This end-to-end automation simplifies the migration process and significantly reduces the potential for manual errors.

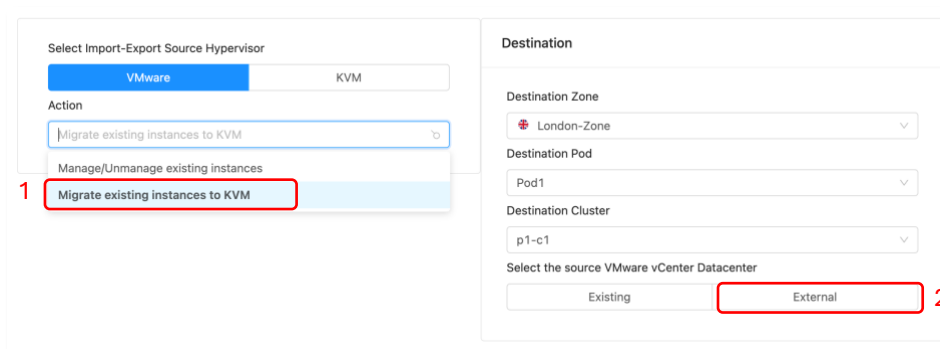
Starting Migrating Instances

It is highly recommended that the Instances to be migrated be turned off during the migration, although it is possible to migrate running Linux Instances, but it is important to consider that the time spent will be longer as it needs to ensure that all data are synchronized to the converted Instance. Windows Instances can only be migrated when turned off.

1. In the CloudStack UI, log in as admin, click on “Tools” > “Import-Export Instances” and then, Select VMware as “Import-Export Source Hypervisor”:



2. In the “Action”, select “Migrate existing Instances to KVM” then, in the “Destination” on the right, choose the “Destination Zone”, “Pod”, the KVM “Cluster” and click on “External”:



3. Insert your “VMware Datacenter vCenter” IP address, the “vCenter Datacenter” name and your credentials (username and password) to connect and then, click on “List VMware Instances”:

Select the source VMware vCenter Datacenter

☐ Existing ☒ External

VMware datacenter vCenter ⓘ

vCenter datacenter ⓘ

vCenter username ⓘ

vCenter password ⓘ

A list of VMware Instances will be shown:

Unmanaged Instances ⓘ ⓘ

Search

Name	State	Host	Cluster name	OS type
☐ Windows-2019-Production	PowerOff	10.0.34.45	cluster-01	Microsoft Windows Server 2016 (64-bit)
☐ AlmaLinux-9-Production	PowerOff	10.0.33.220	cluster-01	Other 5.x Linux (64-bit)
☐ Windows-2019-dev	PowerOff	10.0.33.220	cluster-01	Microsoft Windows Server 2016 (64-bit)
☐ AlmaLinux-9-dev	PowerOff	10.0.34.45	cluster-01	Other 5.x Linux (64-bit)

Showing 1-5 of 5 items < 1 >

4. Select the Instance you want to convert and then, click on “Import Instance”:

Unmanaged Instances ⓘ ⓘ

Search

1 ☒	Windows-2019-Production	PowerOff	10.0.34.45	cluster-01	Microsoft Windows Server 2016 (64-bit)
☐	AlmaLinux-9-Production	PowerOff	10.0.33.220	cluster-01	Other 5.x Linux (64-bit)
☐	Windows-2019-dev	PowerOff	10.0.33.220	cluster-01	Microsoft Windows Server 2016 (64-bit)
☐	AlmaLinux-9-dev	PowerOff	10.0.34.45	cluster-01	Other 5.x Linux (64-bit)

Showing 1-5 of 5 items < 1 >

2

5. A migration form will be open. For each Instance you want to convert, provide the information according to your planning. Following the example of this Guide, for the "Windows-2019-Production" Instance, we will configure it as follows:

Import Instance

X

Unmanaged Instance

 **Windows-2019-Production**

Status

●

 PowerOff

CPU

4 CPU x 0.00 Ghz

Memory

4096 MB memory

Network

1 NIC(s)

eth0

* Display name ⓘ

Windows-2019-Production

Host name ⓘ

windows-2019-production

Domain ⓘ

ROOT/Custom01

Account ⓘ

production

Project ⓘ

import instance for the project

☐ (Optional) Select a KVM host in the cluster to perform the instance conversion through virt-v2v

☐ (Optional) Select a Storage temporary destination for the converted disks through virt-v2v

Compute offering ⓘ

Search

Q

Compute offering	CPU	Memory
☐ Small Instance	1 CPU x 0.50 Ghz	512 MB
☐ Medium Instance	1 CPU x 1.00 Ghz	1024 MB
☐ Large Instance	2 CPU x 1.50 Ghz	2048 MB
☒ X-Large Instance	4 CPU x 1.50 Ghz	4096 MB

Total 4 items < 1 > 10 / page

Network selection ⓘ

IP address changes takes effect only after Instance restart.

Allow duplicate MAC addresses ⓘ

☒

NIC	Network	IP Address
Network adapter 1 mac: 02:01:00:cc:00:02 vlan: 0	Production-Network	☒ Automatically assign a random IP address

Cancel

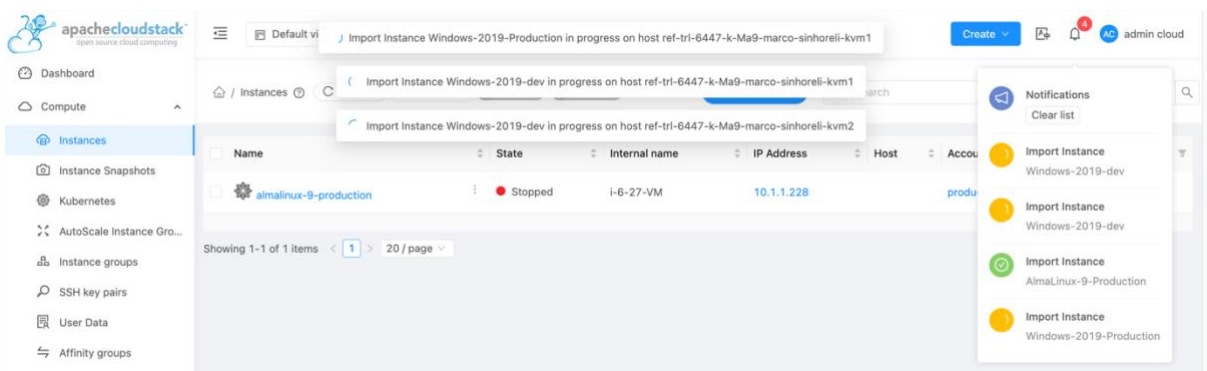
OK

Note that, for the “Windows-2019-Production” Instance, we set it as planned in the example, using the Domain, Account, Compute Offering and Network created before:

Domain: Customer01
Account: development
Compute Offering: X-Large Instance
Network: Production-Network

Tip: Additionally, you can select a different temporary storage to store the converted Instance disk. By default, Secondary Storage will be used. You can also change the converted Instance’s characteristics by selecting a different Compute Offering. Allowing duplicate MAC Addresses will use the same one in the converted Instance and, selecting a specific Host, it will perform the migration in the Hosts you want. Is important to say that, the KVM Host selected need to have installed the virt-v2v tool.

6. In the Notification area, you can monitor the running migration processes. When the conversion is complete, you can see the imported Instances by navigating to “Compute” > “Instances” in the sidebar menu:



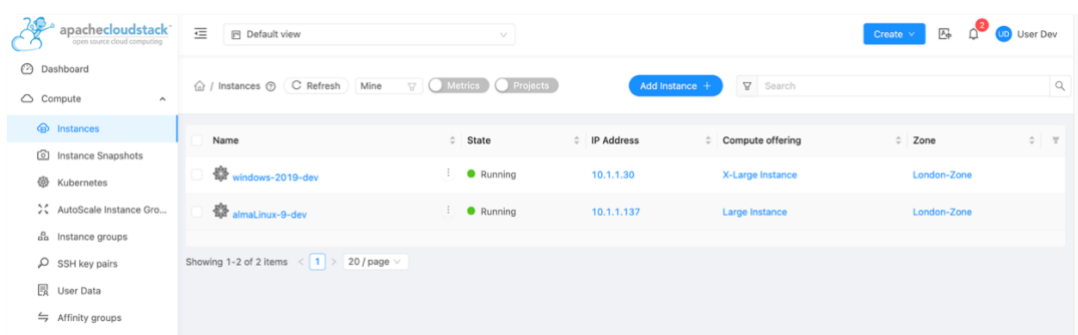
Managing Converted Instances

Once the migration process is concluded, the users are able to manage their Instances in CloudStack. When logged in as a regular user in CloudStack UI, users can manage independently their cloud resources.

- Logged in as dev-user:



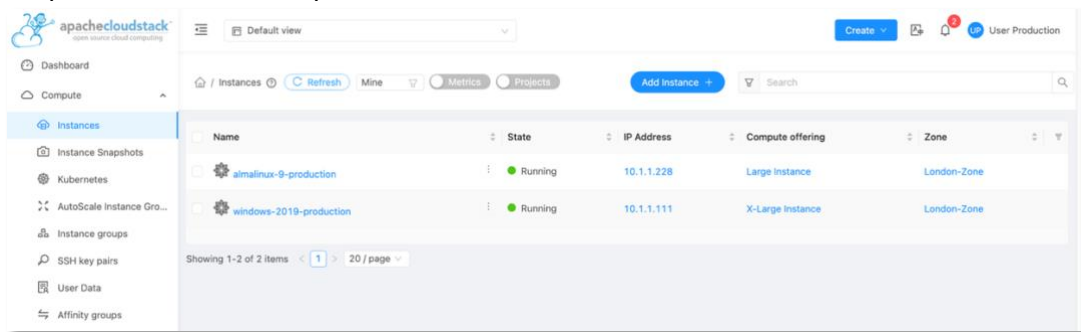
- dev-user Instances panel:



- Logged in as prod-user:



- prod-user Instances panel:



Critical Migration Points

Migrating VMware to Apache CloudStack is a significant change. To ensure a seamless transition and minimise potential disruptions, consider the following critical migration points:

Pre-migration assessment: We recommend doing a full assessment of the existing VMware environment before starting the migration. This includes understanding the technical specifications, dependencies, and operational nuances of all Instances scheduled for migration. This assessment assists in early identification of potential challenges and effective planning to address them.

Migration and phasing strategy: Develop a clear migration strategy that describes the sequence in which Instances will be migrated. Consider starting with non-critical or less complex Instances to refine the migration process. Gradual migration allows lessons learned in early stages to inform strategies and adjustments in subsequent phases, reducing risks and ensuring smoother transitions.

Tools and automation: Take advantage of the automation capabilities of the CloudStack 4.19 migration tool, which simplifies the conversion of VMware instances to CloudStack. Familiarize yourself with its features and limitations to make the most of its features. Verify that all prerequisites, such as the virt-v2v tool and the required drivers, are in place and configured correctly.

Custom scripts and manual adjustments: Although automation tools handle a significant part of the migration process, some scenarios may require custom scripts or manual adjustments. This may involve tweaking settings, optimizing performance after migration, or resolving specific compatibility issues. Planning these contingencies ensures that you can maintain control over the migration process and the outcome.

Testing and validation: After migrating each Instance, conduct testing to ensure that it operates as expected in the new environment. This includes functionality testing, performance benchmarking, and security validations. Establish a clear test protocol to promptly identify and fix any issues, minimising the impact on business operations.

Backup and rollback plans: Despite planning and full testing unexpected problems may arise. Having a robust backup and rollback plan provides a safety net, allowing you to quickly restore operations if needed. This includes not only data backups but also settings and customisations that have been applied.

Training and documentation: The migration process can significantly change the operating scenario. Provide comprehensive training and documentation to your team to ensure they are equipped to manage the new CloudStack environment efficiently. This includes operational procedures, troubleshooting guides, and best practices for using new resources and capabilities.

Post-migration optimization: The migration process does not end with the successful transfer of Instances. After migration, analyse the performance and utilization of your CloudStack environment to identify optimisation opportunities. This may involve tweaking Service Offerings, simplifying Network configurations, or implementing additional CloudStack Global Settings to increase efficiency and performance.

Expert Help: Leverage ShapeBlue Services for Complex Scenarios and Custom Migrations

ShapeBlue is the largest independent integrator of CloudStack technologies globally and are specialists in designing and implementing IaaS cloud infrastructures for both private and public cloud implementations. Our services include consulting, integration, training and infrastructure support. Our technical teams are all active contributors to the Apache CloudStack project, and a good percentage of our team are also project committers.

We are specialised in providing customised support for complex VMware migration scenarios and customised solutions that go beyond standard procedures. With a deep understanding of VMware and Apache CloudStack environments, ShapeBlue's expert teams are equipped to handle complex migration requirements, ensuring a seamless transition with minimal disruption.

Whether you're architecting a custom VMware migration strategy, addressing specific technical challenges, or optimizing your new CloudStack environment, we can offer you access to a wealth of experience and knowledge. This ensures not only a successful migration but also a future-proof, optimised cloud infrastructure aligned with your organisational goals and technical demands.

Automation Options via the CloudStack API

Apache CloudStack provides a Rest API and CLI that allows users and operators to automate any action. This includes all of the processes outlined as part of a VMware migration described in this document. Most organisations undertaking migrations at scale will choose to automate: the entire migration process can be automated end-to-end and repeated for 1000's of VMs

This guide does not cover this automation in detail, but below you see a summary of the API calls typically used in such a migration

API Call	Used for
createDomain	Creates a domain for the different tenants in the VMware environment.
createAccount	Creates accounts for users.
createServiceOffering	Used to create compute offerings of the appropriate size for your VMware VM's.
createNetwork	Used to create networks for the VM's.
importVm	Used to import external VM's as Apache CloudStack instances and convert them from VMware to KVM.

Further information on the Apache CloudStack API can be found at [CloudStack API Reference \(apache.org\)](#).



Need help with **Apache CloudStack**?

Get in Touch



 cloudstack.apache.org/

About **Apache CloudStack**

Apache CloudStack is the leading open source cloud orchestration platform, in use by many of the world's largest public and private clouds. It is a multi-hypervisor, multi-tenant, high-availability Infrastructure as a Service cloud management platform. CloudStack is software that provides a cloud orchestration layer, giving automation of the creation, provisioning and configuration of IaaS components.

CloudStack turns an existing virtual infrastructure into a cloud-based infrastructure as a Service (IaaS) platform. The fact CloudStack leverages existing infrastructure means that the cost and time for an organisation to build a multi-tenant IaaS platform is greatly reduced.



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 www.shapeblue.com

About **ShapeBlue**

ShapeBlue is the largest independent integrator of CloudStack technologies globally and are specialists in the design and implementation of IaaS cloud infrastructures for both private and public cloud implementations. We combine 100's of person-years of experience in designing and building complex network, storage and compute infrastructures with globally leading skills in Apache CloudStack.